MONTE CARLO MODELLING OF FERROELECTRIC POLARIZATION FOR NEUROMORPHIC APPLICATIONS

Stochastic Polarization Dynamics in Polycrystalline HfO_2 for FeFET-Based Neuromorphic Computing

Johnson Jasson
Advised by:
Prof. A N M Nafiul Islam

Department of Mechanical, Electrical and Renewable Engineering

INTRODUCTION

- **Polycrystalline HfO_2** ferroelectrics exhibit **stochastic polarization** switching due to variations in **domain** activation energies and switching time constants, resulting in **history-dependent** behavior.

- **Monte Carlo simulation** is employed, (Python, MATLAB), to model domain-level polarization dynamics and investigate switching mechanisms in ferroelectric capacitors and FeFETs.

- The **intrinsic memory** of ferroelectric polarization *resembles* **biological synaptic plasticity**, making FeFETs promising candidates for **efficient neuromorphic computing**.

THEORY & MODEL

1 Switching Time Constant

$$\tau = \tau_\infty \exp[(Ea/E)^\alpha]$$

2 History Parameter

$$h_{new} = h_{old} + \Delta t/\tau$$

3 Switching Probability

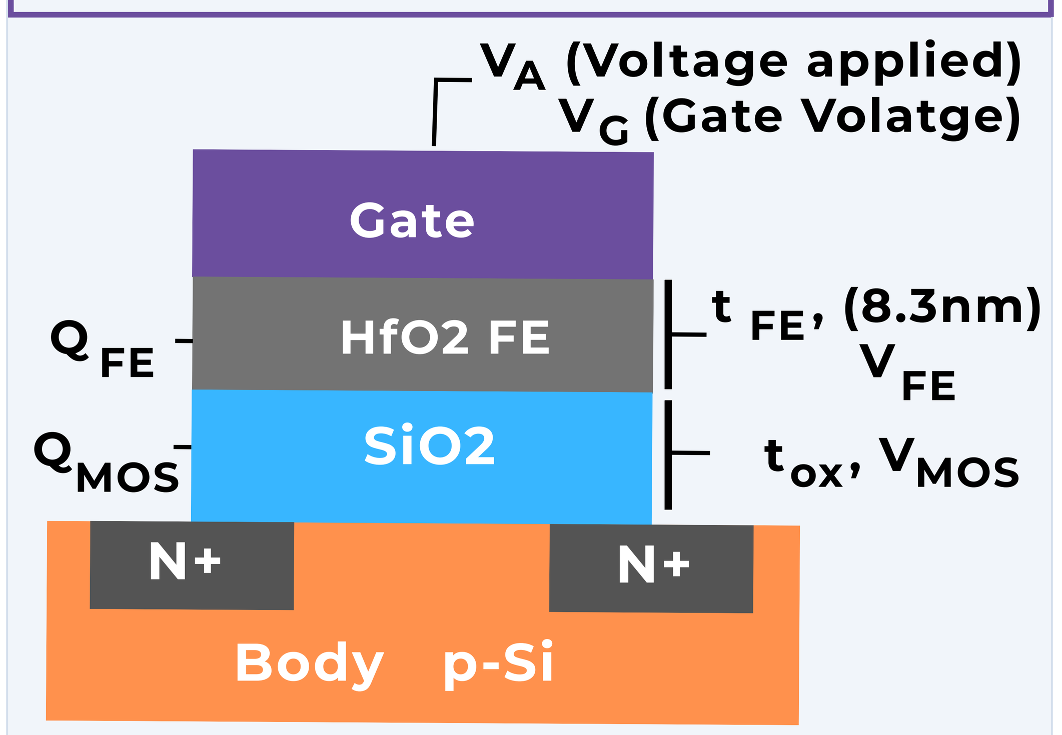
$$P_{sw} = 1 - \exp[h_{old}^{\beta} - h_{new}^{\beta}]$$

4 Polarization

$$P_{FE}(t) = P_R (1/N) \sum s_i(t) \quad (i = 1 \ldots N)$$

5 FeFET Charge Equation

$$Q_{MOS}(V_G - V_{FE}) = P_{FE} + C_{FE} V_{FE}$$



FRAMEWORK

Applied Voltage Pulse $V_A(t)$

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Monte Carlo Domain Switching

↓

Polarization P_{FE} (Ferroelectric Polarization)

↓

Self-Consistent FeFET Solver ($Q_{MOS} = Q_{FE}$)
$V_G = V_{FE} + V_{MOS}$ • Charge balance

↓

V_{FE} and V_{MOS}

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Surface-Potential MOS Model

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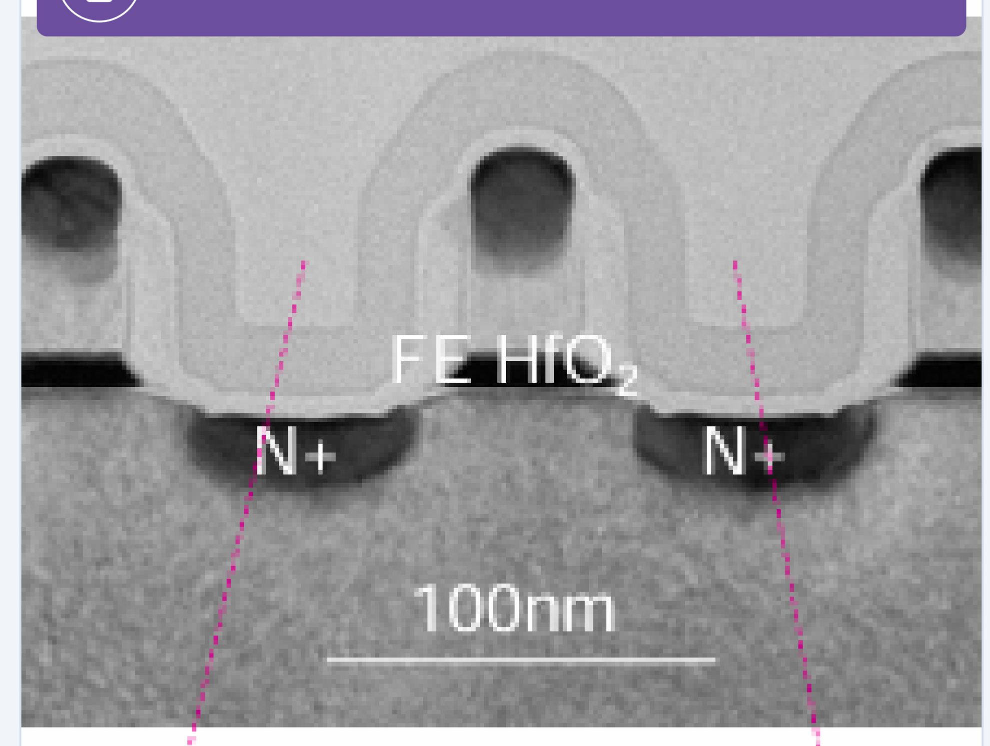
Drain Current, I_D

↓

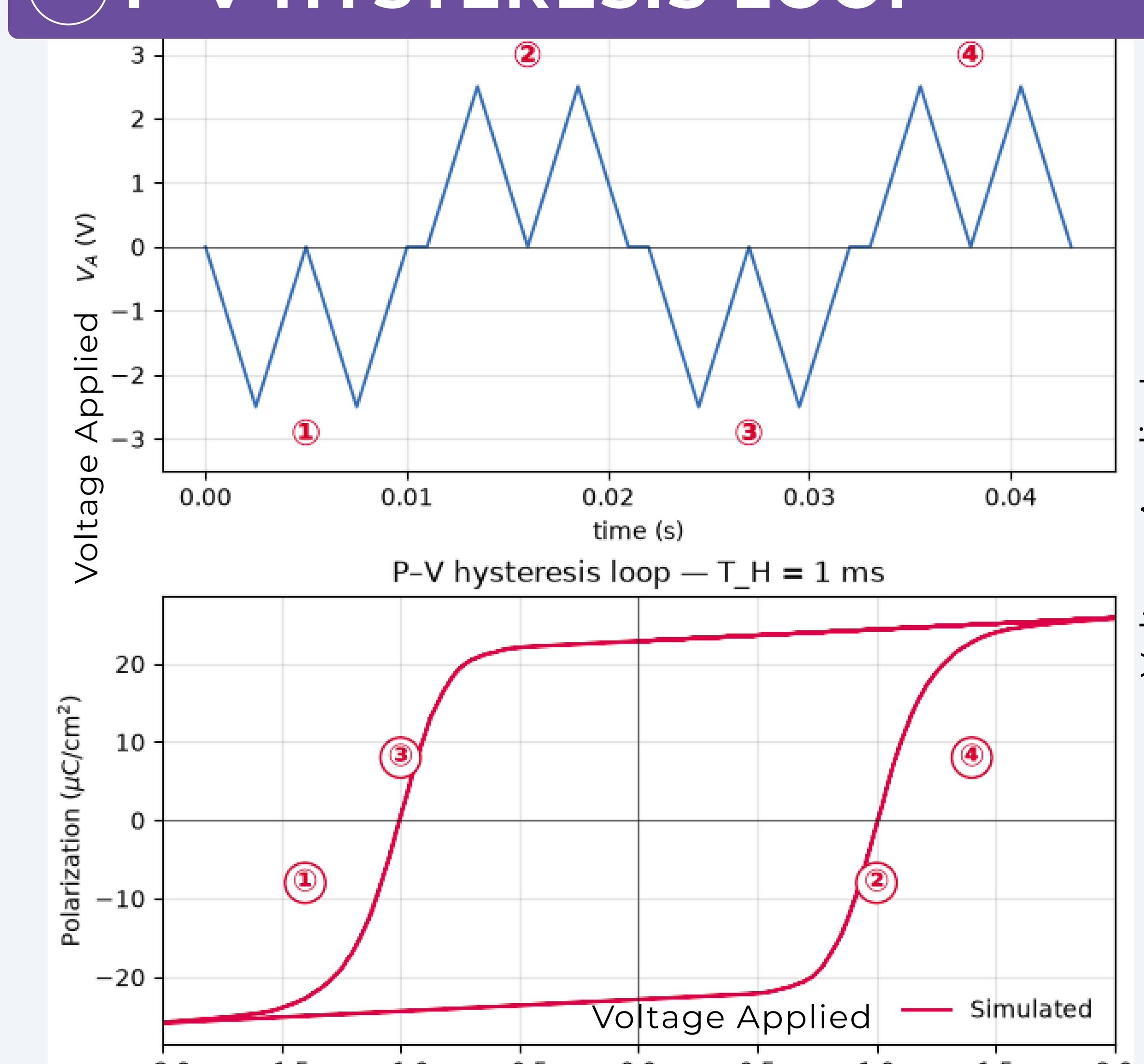
Device-Response (Neuromorphic Behavior)

FeFET DEVICE STRUCTURE

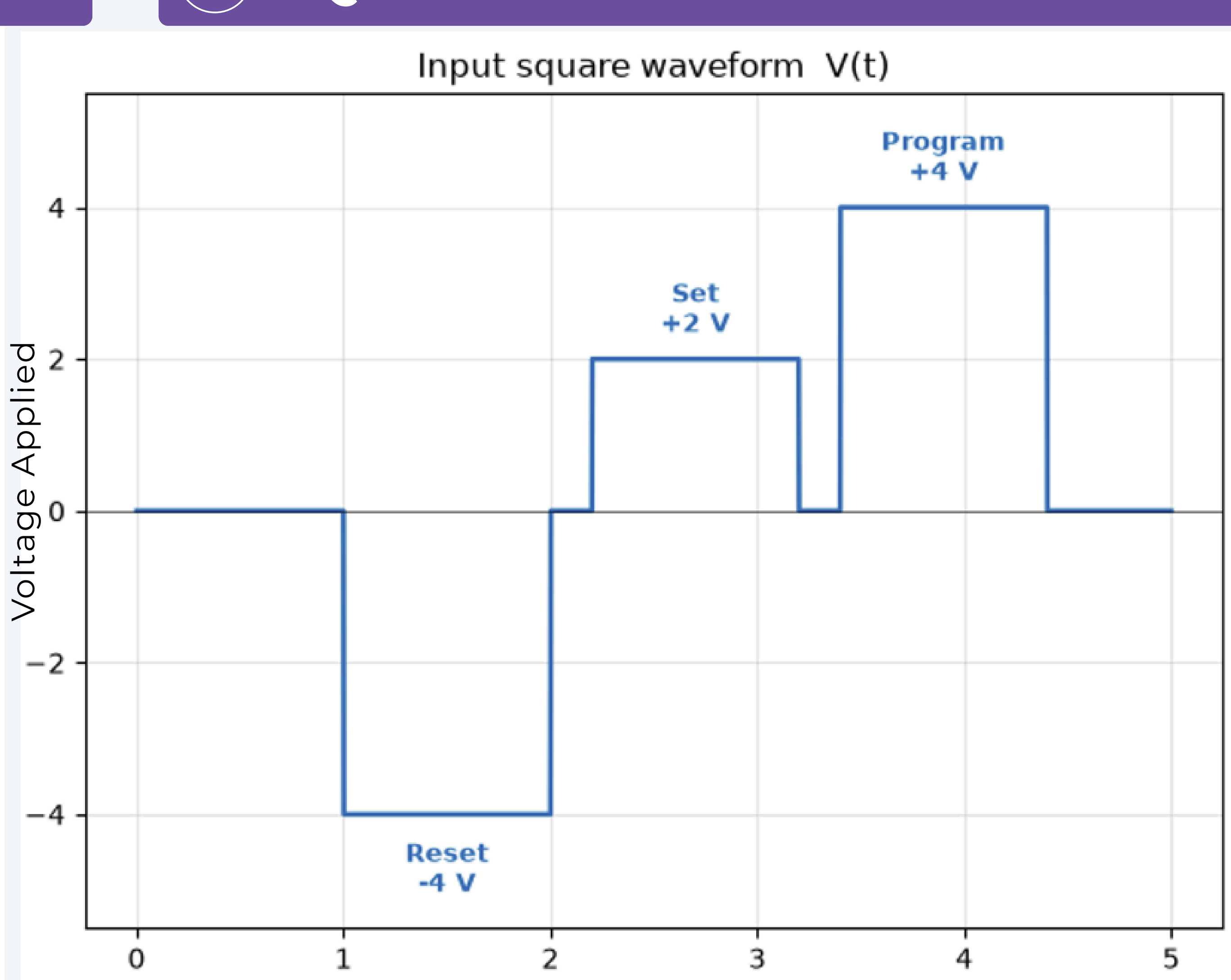


Ferroelectric Material	HfO_2
Ferroelectric Thickness (t_{FE})	8.3 nm
Gate Oxide (t_{OX})	1.5 nm SiO_2
P_R (Remanent Polarization)	22.9 $\mu C/cm^2$
Vmax (Pulse Amplitude)	±4 V
Monte Carlo Grains (N)	5000
crystallinity	Polycrystalline

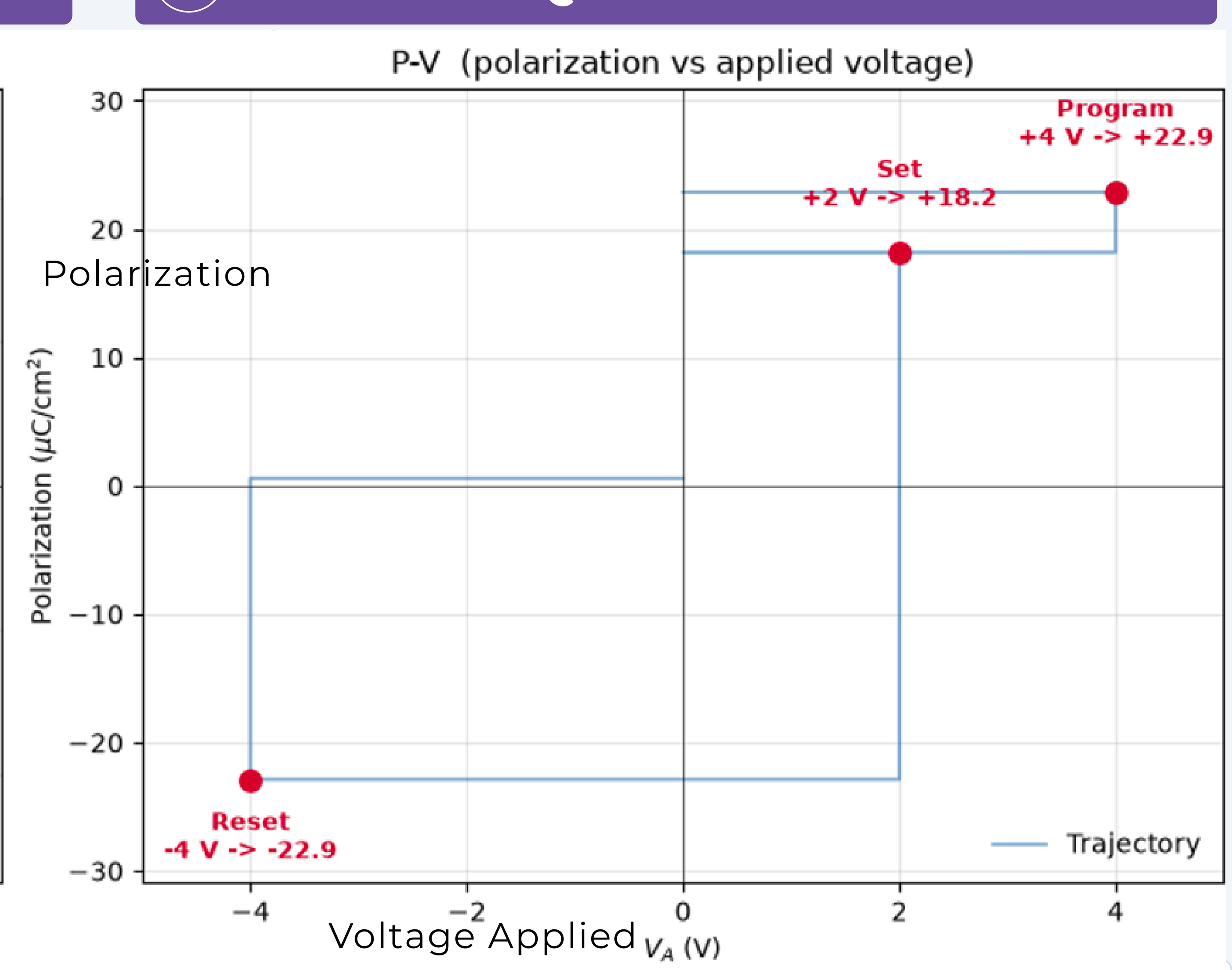
P-V HYSTERESIS LOOP



SQUARE WAVEFORM



PV FOR SQUARE WAVEFROM

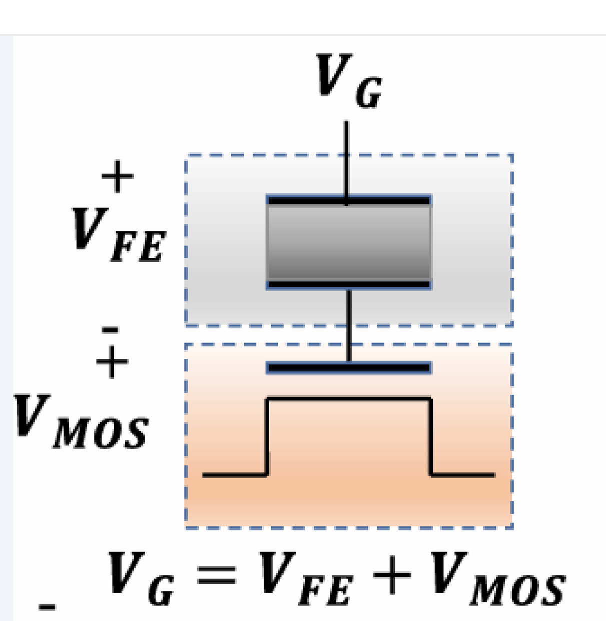


RESULTS & DISCUSSION

- Polarization (P_{FE}) depends on the **pulse duration** and **magnitude** of the V_G

- The framework enables accurate prediction of FeFET behavior for neuromorphic applications.

τ = Domain switching time
Ea = Activation electric field
E = Applied electric field
α, β = Empirical exponents (Weibull)
h = history parameter
P_{SW} = Probability of switching
P_R = Remanent polarization
P = Polarization, N = No. of grains
S_i = domain state (+1, -1)
C_{FE}, V_{FE} = Ferroelectric capacitance, voltage
Δt = time step, **t** = instantaneous time



CONCLUSION

- The **model** captures **stochastic** switching, hysteresis behavior, and self-consistent electrostatics, forming a robust basis for designing FeFETs for **efficient** neuromorphic computing.

- Partial polarization switching mimics **synaptic plasticity**, making FeFETs suitable in neuromorphic hardware

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